

KOCHI, Japan

WMO: 478930

Lat: 33.5683N

Lon: 133.5490E

Elev: 5

StdP: 101.27

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.9	0.2	-10.1	1.6	6.0	-8.5	1.8	6.2	6.3	8.7	5.5	8.8	1.7	280	0.331

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.8	33.1	25.5	32.2	25.3	31.3	25.1	26.8	30.5	26.4	30.0	26.0	29.5	2.8	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.9	21.2	28.7	25.4	20.6	28.4	25.0	20.1	28.2	84.4	30.7	82.4	30.1	80.7	29.5	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.3	4.4	3.9	-2.8	35.5	1.1	1.0	-3.7	36.2	-4.3	36.8	-5.0	37.3	-5.8	38.0		
(5)			-4.3	27.6	1.1	0.8	-5.1	28.2	-5.8	28.6	-6.4	29.0	-7.2	29.6		

Monthly Climatic Design Conditions

			Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
			(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	17.6	6.9	8.1	11.2	15.8	20.1	23.3	27.3	28.2	25.4	20.3	14.5	9.2	(6)
(7)		DBStd	7.74	2.50	3.14	2.98	2.79	1.91	2.00	1.79	1.40	2.16	2.69	3.23	2.98	(7)
(8)		HDD10.0	249	101	71	22	1	0	0	0	0	0	0	3	52	(8)
(9)		HDD18.3	1367	354	288	222	83	6	0	0	0	0	11	119	284	(9)
(10)		CDD10.0	3012	5	17	58	174	314	398	536	564	462	321	139	26	(10)
(11)		CDD18.3	1089	0	0	0	7	62	148	277	306	212	73	5	0	(11)
(12)		CDH23.3	9487	0	0	1	19	231	821	2867	3474	1729	336	9	0	(12)
(13)		CDH26.7	3056	0	0	0	1	16	122	998	1382	502	36	0	0	(13)
(14)	Wind	WSAvg	1.8	1.8	1.8	1.9	1.9	1.8	1.6	1.8	1.8	1.8	1.7	1.7	1.7	(14)
(15)	Precipitation	PrecAvg	2653	61	97	172	267	260	377	306	319	356	180	134	63	(15)
(16)		PrecMax	4383	195	332	302	534	516	862	875	1561	1355	580	415	315	(16)
(17)		PrecMin	1739	0	21	16	86	49	119	41	12	57	24	8	0	(17)
(18)		PrecStd	573	44	72	73	104	98	158	183	260	243	117	100	55	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.6	19.6	21.6	25.4	28.4	31.0	34.2	34.6	32.8	29.1	24.4	19.8	(19)
(20)			MCWB	10.6	14.5	14.2	16.9	18.2	24.5	25.4	25.3	24.5	21.7	17.7	14.1	(20)
(21)		2%	DB	14.6	17.2	19.7	23.6	26.7	29.2	32.8	33.4	31.5	27.6	22.8	17.7	(21)
(22)			MCWB	9.2	12.7	13.4	16.0	18.4	23.1	25.5	25.6	24.1	20.7	16.7	11.8	(22)
(23)		5%	DB	13.3	15.4	18.3	22.2	25.6	28.0	31.9	32.6	30.5	26.5	21.4	16.1	(23)
(24)			MCWB	8.0	10.6	12.7	15.7	18.3	22.5	25.4	25.5	23.7	19.9	15.8	10.8	(24)
(25)		10%	DB	12.0	13.7	16.8	21.1	24.6	27.0	31.0	31.7	29.5	25.3	20.0	14.5	(25)
(26)			MCWB	7.2	8.8	11.8	15.2	18.1	21.9	25.1	25.4	23.4	19.1	15.1	9.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.3	16.4	17.1	20.5	22.9	25.9	27.2	27.5	26.7	24.9	20.6	16.4	(27)
(28)			MCDWB	14.7	18.2	18.8	22.5	24.6	28.9	31.3	31.4	29.9	27.0	21.9	18.0	(28)
(29)		2%	WB	11.0	14.1	15.5	19.0	21.9	25.2	26.6	27.0	26.0	23.3	18.8	13.6	(29)
(30)			MCDWB	12.6	16.2	17.5	21.2	23.7	27.6	30.3	30.8	29.0	25.8	20.7	15.9	(30)
(31)		5%	WB	9.3	11.8	14.3	17.9	21.0	24.4	26.2	26.6	25.5	21.9	17.4	12.0	(31)
(32)			MCDWB	11.8	14.1	16.9	20.3	23.0	26.7	29.7	30.3	28.5	24.4	19.8	14.3	(32)
(33)		10%	WB	7.8	10.0	13.0	16.9	19.9	23.6	25.8	26.2	24.9	20.8	16.2	10.6	(33)
(34)			MCDWB	11.2	12.7	15.8	19.6	22.7	25.7	29.2	29.9	27.9	23.5	18.9	13.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.3	9.2	9.2	9.1	8.4	6.6	6.2	6.8	7.4	8.5	9.0	9.3	(35)
(36)			MCDBR	10.6	10.1	10.5	10.3	10.1	7.9	7.8	8.0	8.2	9.3	9.5	10.5	(36)
(37)			MCWBR	6.4	6.4	6.7	5.7	5.0	3.5	3.0	3.0	3.4	4.3	5.2	6.2	(37)
(38)		5% WB	MCDBR	8.7	8.3	8.5	7.5	6.9	5.6	6.5	6.6	6.4	6.6	7.2	8.3	(38)
(39)			MCWBR	6.5	6.5	6.8	5.3	4.8	2.9	2.8	2.8	3.1	4.0	5.0	6.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.424	0.479	0.547	0.578	0.582	0.596	0.591	0.560	0.520	0.480	0.443	0.414	(40)	
(41)		taud	2.077	1.959	1.811	1.792	1.845	1.885	1.940	2.011	2.087	2.109	2.116	2.121	(41)	
(42)		Ebn at Noon	755	752	733	735	737	724	725	740	751	745	730	739	(42)	
(43)		Edh at Noon	130	163	204	217	208	199	188	172	153	139	125	119	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.65	3.21	4.14	4.83	5.10	4.31	4.93	5.04	4.04	3.41	2.66	2.31	(44)	
(45)		RadStd	0.21	0.24	0.31	0.53	0.56	0.54	0.61	0.52	0.39	0.36	0.24	0.15	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	+0.37	+0.51	N/A	N/A	N/A	N/A
(47) Regional (23 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page